PEARSON EDEXCEL INTERNATIONAL GCSE

Mathematics A

Unit 2 • Chapter OL-T16 Classified

Cross-session • chapter-organised candidate

6 classified items • 22 marks • 1 chapter(s)

Source-faithful practice with synchronized solutions

Every question is followed by its worked answer/reference

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Contents

Unit 2 • Unit 2 • Chapter OL-T16 • 6 items • 22 marks

June 2026 is intentionally deferred; November 2025 remains provisional QP-only in this private candidate.

Algebra Toolkit

Unit 2 • 22 marks • 6 item(s)

Original question context and synchronized companion pages follow.

Terminal-demand ownership is preserved; prerequisite links remain in the internal ledger.

Question 20

4MA1-2H Q15

ORIGINAL QUESTION

4 marks

15 (a) Make g the subject of $e = \sqrt{\dfrac{7g + 5}{11 + 2g}}$

....

(4)

Worked Solution 20

Change the subject

MODULAR UNIT 2
4 marks

Square and remove the fraction

$$e^2(11 + 2g) = 7g + 5.$$

Collect the terms in the new subject

$$g(2e^2 - 7) = 5 - 11e^2 \Rightarrow g = \frac{5 - 11e^2}{2e^2 - 7}.$$

Final answer

$$g = \frac{5 - 11e^2}{2e^2 - 7}$$

Question 23

4MA1-2H Q20

ORIGINAL QUESTION

3 marks

20 Given that $k = x - y$ and $x = \dfrac{1}{4y}$

express $\dfrac{5k}{x+2}$ in the form $\dfrac{a - by^2}{c + dy}$ where a, b, c and d are integers.

....

Worked Solution 23

Algebraic substitution

MODULAR UNIT 2

3 marks

Substitute

$$x = 1/(4y), \quad k = 1/(4y) - y.$$

Simplify

$$\frac{5k}{x+2} = \frac{5-20y^2}{1+8y}.$$

Final answer

$$\frac{5-20y^2}{1+8y}$$

Question 12

4MA1-1H Q15 demand

ORIGINAL QUESTION UNIT 2

3 marks

15 Make n the subject of the formula $x = \dfrac{3p+n}{3n-4}$

....

Worked solution 12

Make n the subject

MODULAR UNIT 2
3 marks

Demand – Make n the subject

Clear the denominator

$$x(3n-4)=3p+n \Rightarrow 3nx-4x=3p+n.$$

Collect n terms

$$n(3x-1)=3p+4x.$$

Divide

$$n=\frac{3p+4x}{3x-1}.$$

Final answer

n = (3p + 4x)/(3x - 1)

Question 27

4MA1-2H Q17

ORIGINAL QUESTION
4 marks

(b) Make e the subject of $w = \sqrt{\dfrac{e+g}{ef-d}}$

....

(4)

Worked Solution 27

Make e the subject of a square-root formula

MODULAR UNIT 2
4 marks

Square

$$w^2 = \frac{e+g}{ef-d}.$$

Clear the denominator

$$w^2(ef-d) = e+g \Rightarrow w^2ef - w^2d = e+g.$$

Collect e

$$e(w^2f-1) = g + w^2d.$$

Divide

$$e = \frac{g+w^2d}{w^2f-1}.$$

Final answer

$$e = \frac{g+w^2d}{w^2f-1}$$

Original question 26

4MA1-2H Q12 M02

MODULAR UNIT 2

4 marks

(b) Make c the subject of the formula $f = \sqrt{\dfrac{a + bc}{c - d}}$

....

(4)

Worked solution 26
Make c the subject

MODULAR UNIT 2
4 marks

M02 – Make c the subject

Square

$$f^2 = (a + bc)/(c - d).$$

Multiply out

$$cf^2 - df^2 = a + bc.$$

Collect c

$$c(f^2 - b) = a + df^2.$$

Divide

$$c = (a + df^2)/(f^2 - b).$$

Final answer

$$c = \frac{a + df^2}{f^2 - b}$$

Original question 27

4MA1-2H Q14 Q14(b)

ORIGINAL QUESTION
4 marks

(b) Make c the subject of $g = \dfrac{c+3}{4+c} - 7$

....

(4)

Worked solution 27

Chapter: Algebra Toolkit

MODULAR UNIT 2
4 marks

Q14(b) – Chapter: Algebra Toolkit

Family: Change a subject that appears twice or as a power • Variant: Subject appears in two linear terms

Method
$g + 7 = (c + 3)/(c + 4)$.
Why: Add 7 to both sides.

Method
$(g + 7)(c + 4) = c + 3$, so $gc + 7c + 4g + 28 = c + 3$.
Why: Remove the fraction and expand.

Method
$c(g + 6) = -(4g + 25)$.
Why: Collect all c-terms on one side.

Method
Therefore $c = -(4g + 25)/(g + 6)$.
Why: Divide by the coefficient of c.

Final answer
c: $c = -(4g + 25)/(g + 6)$